**Directions** *(Questions 505-506): These questions are based on the following information:*

In a holy city, there are ten shrines and a certain number of holy lakes. A group of pilgrims stayed in the city for a few days and visited the shrines and lakes during their stay. At the end of the stay it turned out that in all each shrine was visited exactly by 4 pilgrims and each lake was visited exactly by 6 pilgrims. Each pilgrim visited exactly 5 shrines and 3 lakes.

**505.** The number of lakes in the holy city is

(*a*) 4 (*b*) 6
(*c*) 8 (*d*) 10

**506.** The number of pilgrims in the group is

(*a*) 4 (*b*) 6
(*c*) 8 (*d*) 10

**507.** From a number of apples, a man sells half the number of existing apples plus 1 to the first customer, sells $\frac{1}{3}$rd of the remaining apples plus 1 to the second customer and $\frac{1}{5}$th of the remaining apples plus 1 to the third customer. He then finds that he has 3 apples left. How many apples did he have originally?

(*a*) 15 (*b*) 18
(*c*) 20 (*d*) 25

**508.** Ravi has two examinations on Wednesday – Engineering Mathematics in the morning and Engineering Drawing in the afternoon. He has a fixed amount of time to read the textbooks of both these subjects on Tuesday. During this time he can read 80 pages of Engineering Mathematics and 100 pages of Engineering Drawing. Alternatively, he can also read 50 pages of Engineering Mathematics and 250 pages of Engineering Drawing. Assume that the amount of time it takes to read one page of the textbook of either subject is constant. Ravi is confident about Engineering Drawing and wants to devote full time to reading Engineering Mathematics. The number of Engineering Mathematics text book pages he can read on Tuesday is **(M.B.A., 2007)**

(*a*) 60 (*b*) 100
(*c*) 300 (*d*) 500

**509.** Rohan went to the post-office to buy five-rupee, two-rupee and one-rupee stamps. He paid the clerk ₹ 20, and since he had no change, he gave Rohan three more one-rupee stamps. If the number of stamps of each type that he had ordered initially was more than one, what was the total number of stamps that he bought?

(*a*) 8 (*b*) 9
(*c*) 10 (*d*) 12

**510.** Aditya went to a stationery shop to buy some Parker pens. Gel pens cost ₹ 300 each while fountain pens cost ₹ 400 each. Aditya spent a total of ₹ 3600 on pens. If he had bought as many fountain pens as the number of gel pens he actually bought and vice versa, he would have saved an amount equal to half the cost of one pen of one of the two types. Find the total number of pens he bought.

(*a*) 8 (*b*) 9
(*c*) 10 (*d*) 12

**Directions** *(Questions 511-513): Study the following information carefully to answer these questions:*

A young girl Roopa leaves home with $x$ flowers and goes to the bank of a nearby river. On the bank of the river, there are four places of worship, standing in a row. She dips all the $x$ flowers into the river, the number of flowers doubles. Then, she enters the first place of worship and offers $y$ flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the second place of worship and offers $y$ flowers to the deity. She dips the remaining flowers into the river and again the number of flowers doubles. She goes to the third place of worship and offers $y$ flowers to the deity. She dips the remaining flowers into the river and again the number of flowers doubles. She goes to the fourth place of worship and offers $y$ flowers to the deity. Now she is left with no flowers in hand.

**511.** If Roopa leaves home with 30 flowers, the number of flowers she offers to each deity is

(*a*) 30 (*b*) 31
(*c*) 32 (*d*) 33

**512.** The minimum number of flowers that could be offered to each deity is

(*a*) 0 (*b*) 15
(*c*) 16 (*d*) Cannot be determined

**513.** The minimum number of flowers with which Roopa leaves home is

(*a*) 0 (*b*) 15
(*c*) 16 (*d*) Cannot be determined

**514.** The owner of a local jewellery store hired 3 watchmen to guard his diamonds, but a thief still got in and stole some diamonds. On the way out the thief met each watchman, one at a time. To each he gave $\frac{1}{2}$ of the diamonds he had then, and 2 more besides. He escaped with one diamond. How many did he steal originally?

(*a*) 25 (*b*) 36
(*c*) 40 (*d*) None of these

**515.** Three friends returning from a movie, stopped to eat at a restaurant. After dinner, they paid their bill